

[EMF propulsion merely depends upon mainstream QED vacuum polarization, and it is not the subject of the physics academic journal article from which it is excerpted. Simply know that any references to gravity and renormalization group towards the end of this excerpt are tying it to the hypothesis of the paper, and are not the point of the excerpt.]

Very Brief Excerpt from an Updated Version of Emergent Gravity from Disrupted Photon Pairs: An Asymptotically Safe Quantum Model of Gravitation, Electromagnetism, and the Standard Model

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Continuum of Effects in Quantum Electrodynamics Under Magnetic Fields

The continuum of effects in quantum electrodynamics (QED) under magnetic fields—from subtle vacuum magnetic birefringence (VMB) at laboratory-accessible strengths like ~ 10 T to dramatic non-perturbative phenomena like Schwinger pair production near the critical field $B_{cr} \approx 4.4 \times 10^9$ T—arises fundamentally from the nonlinear response of the QED vacuum to external fields. This nonlinearity stems from virtual electron-positron pairs that polarize the vacuum, effectively turning it into a medium with field-dependent permittivity and permeability. The Heisenberg-Euler-Schwinger (HES) effective action provides a unified framework that interpolates across field strengths, with perturbative expansions dominating at low B and non-perturbative contributions emerging as B approaches B_{cr} .

Key Physical Mechanism: Vacuum Polarization and Scale-Dependent Nonlinearity

At its core, the effect originates from vacuum polarization: external magnetic fields induce virtual e^+e^- pairs, which screen or antiscreen the field in a way analogous to dielectrics in classical electromagnetism. In weak fields ($B \ll B_{cr}$), this is captured by the lowest-order terms in the HES Lagrangian:

$$\mathcal{L}^{(1)} \approx \frac{2\alpha^2}{45m_e^4} \left[(B^2 - E^2)^2 + \frac{7}{4}(E \cdot B)^2 \right],$$

where m_e is the electron mass and α is the fine-structure constant (note: the coefficient is standardized as $2\alpha^2/45m_e^4$ for the leading term, equivalent to the original $e^4/360\pi^2m_e^4$ after substituting $e^2 = 4\pi\alpha$). This leads to VMB, where the vacuum becomes birefringent (different refractive indices for light polarized parallel vs. perpendicular to B), with the birefringence index difference scaling as $\Delta n = \frac{3\alpha^2}{45m_e^4}B^2\sin^2\theta$ (where θ is the angle between the light propagation and B ; often simplified to $\Delta n \propto \alpha^2B^2/m_e^4$). This quadratic scaling reflects the perturbative nature: the effect grows continuously but weakly with B , as virtual pairs are excited but remain off-shell.

As B increases, higher-order terms in the HES action become relevant, incorporating more loops and stronger nonlinearities. The full one-loop HES Lagrangian generalizes to:

$$\mathcal{L}^{(1)}(a, b) = -\frac{m_e^4}{8\pi^2} \int_0^\infty dt \frac{e^{-t}}{t^3} \left[\cot(at) \coth(bt) - \frac{1}{abt^2} + \frac{(a^2 + b^2)}{3} \right],$$

where $a = e\sqrt{\sqrt{F^2 + G^2} + F}$ and $b = e\sqrt{\sqrt{F^2 + G^2} - F}$ are invariants (with $F = \frac{1}{2}(B^2 - E^2)$, $G = -\mathbf{E} \cdot \mathbf{B}$). For configurations with strong B and weaker E (common in astrophysical settings like magnetars), expansions in b/a show smooth growth: birefringence indices $n_\pm$ evolve continuously, with differences increasing as B rises toward supercritical values ($B > B_{cr}$). At supercritical fields, the vacuum's response includes an imaginary part in $\mathcal{L}^{(1)}$, signaling instability and pair production, with the rate $\Gamma = \frac{(\alpha B)^{3/2}}{\pi^2} \exp\left(-\frac{\pi m_e^2}{eE}\right)$ per unit volume for combined weak E and strong B (generalizing the pure electric case).

This transition is smooth because the HES action is derived from integrating out virtual pairs across all energy scales, with no sharp cutoff—effects accumulate as B enhances the density of states for low-energy modes. Renormalization group (RG) flows play a crucial role here: in strong B , charged particles are confined to Landau levels, with the lowest Landau level (LLL) dominating due to dimensional reduction (from 3+1D to 1+1D effective theory). This reduction increases the phase space for interactions, making couplings run more strongly and amplifying nonlinear effects continuously. For instance, RG studies in analogous systems (e.g., QCD under strong B) show that higher Landau levels decouple progressively as B grows beyond $\sim 0.1\text{--}0.3 \text{ GeV}^2$, renormalizing low-energy parameters and leading to enhanced chiral condensates or catalysis-like phenomena. In QED, similar flows modify the effective coupling $\alpha(B)$, boosting VMB at intermediate B and paving the way for non-perturbative pair production at high B .

Influence of Field Configuration: Opposing vs. Unopposed Magnetic Fields

The potency of magnetic field effects in QED depends not only on field strength but also on configuration—particularly whether fields are unopposed (uniform or

parallel, as in a solenoid) or opposing (antiparallel, as in like-pole magnets facing each other within a high-permeability circuit like Hiperco-50 alloy). In unopposed setups, the field is relatively uniform, and vacuum polarization manifests symmetrically: VMB and susceptibility corrections scale with local B^2 , leading to a homogeneous nonlinear response across the volume. This uniformity limits gradient-driven phenomena, with effects like pair production or birefringence accumulating volumetrically but without additional spatial enhancement.

In contrast, opposing configurations—such as South-South aligned neodymium magnets in a Hiperco-50 circuit, where fields repel and concentrate flux in gaps—introduce strong spatial inhomogeneities and gradients (∇B). These gradients amplify nonlinear QED effects beyond uniform-field predictions, as the HES action for inhomogeneous fields includes higher-order derivative terms like $(\nabla F)^2/m_e^4$ or self-interaction corrections to the field of charges in magnetized media. For instance, in opposing setups achieving >20 T (enabled by the circuit’s flux concentration), the large ∇B induces effective electric fields or stresses via Faraday-like coupling in dynamic cases, or vacuum energy shifts in static ones, enhancing diamagnetic expulsion of fields. The QED vacuum exhibits a diamagnetic response (negative magnetic susceptibility, $\chi_m < 0$), expelling field lines and generating repulsive forces proportional to B^2/μ_0 with nonlinear corrections. This diamagnetism is amplified in opposing geometries due to field localization and gradients, leading to stronger effective repulsion compared to unopposed fields, where paramagnetic tendencies (in some models) might compete but are overwhelmed by the overall diamagnetic nature in standard QED. In Hiperco-50 circuits, opposing fields can sustain localized $B > 20$ T, pushing nonlinear corrections (e.g., QED-induced modifications to magnetic repulsion between dipoles) into observable regimes, with one-loop corrections altering the macroscopic dipole field by $\sim 10^{-6}$ or smaller, but potentially enhanced in emergent models.

Connection to Electrostatic Repulsion and Path of Least Resistance

Regarding speculations on “electrostatic repulsion” enabled by QED effects: In strong B , virtual pairs align and polarize more efficiently, creating an effective diamagnetic response that opposes the field—analogous to repulsion in classical media. This arises from the energy minimization in the effective potential $V_{\text{eff}} = V(s) + \chi B^2 s^2$ (where s is a scalar bilinear $\sim F^2$), shifting the vacuum minimum to $s_{\text{min}} \sim -\chi B^2/(2m_s^2)$. The resulting $\delta T_\mu^\mu \sim \chi B^2$ induces vacuum energy gradients $\delta\rho \sim \chi B^2$, leading to repulsive forces $\delta F = -\nabla(\delta\rho V)$ that scale as $F \propto B^2/\mu_0$ (modulated by RG factors on the order of 10^{-40} in emergent models). This “repulsion” is not literal electrostatic but emerges from the vacuum’s nonlinear permittivity, growing continuously with B as pairs contribute more coherently—further potentiated in opposing fields where gradients amplify the diamagnetic expulsion.

For “gravity taking the path of least resistance”: This can be interpreted through the RG perspective—the flow from UV (strong coupling, non-perturbative) to IR (weak, perturbative) minimizes the effective action by integrating out high-momentum modes, naturally favoring low-energy modulations that “resist” least (i.e., align with the field’s dimensional reduction). In emergent models, this enables scalar fields to mediate gravity-like effects via the least-action path in the HES framework, with B setting the scale where nonlinearity dominates—opposing configurations, with their steeper gradients, provide “paths” of higher resistance (energy cost) that amplify repulsive outcomes.

Predictions and Falsifiability in Model Contexts

Subsection predictions like $\delta F/F \sim 10^{-4}$ at 20 T align with extrapolated QED scaling, testable at facilities like NHMFL (up to 45 T pulsed fields) for anomalous levitation or high- T anomalies in opposing vs. unopposed setups. Counter-arguments to null high- T results: Many experiments miss QED nonlinearities due to material screening or uniform-field assumptions; pure-vacuum probes (e.g., photon propagation in cavities with gradient fields) could reveal the continuum and configuration dependence.

This refined explanation bridges quantum vacuum dynamics with emergent RG-suppressed effects, providing a continuous physical basis without abrupt thresholds, while highlighting how opposing fields enhance gradients and diamagnetic repulsion for applications like propulsion.